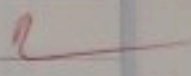
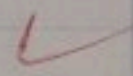
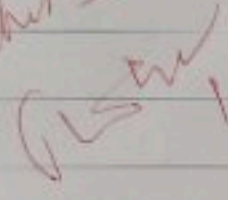


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S. No.	Date	Name of the Experiment	Page No.	Marks Awarded	Signature
1	21/1/2026	Verification of Kirchhoff's voltage law and Kirchhoff's current law.		100	
2	28/1/2026	open circuit and load characteristic of DC shunt generator		100	
03.	04/2/2026	Staircase wiring.		99	
04.	18/02/26	Half wave and Full wave Rectifier		99	
05.	25/2/26	Zener diode as voltage Regulation		99	
06.	11/03/26	Characteristic of LVDT (characteristic variable different Transformer)		97	
Completed  11/03/26					

Half wave and Full wave Rectifier

Aim:

To Simulate the following circuit protocols
i) Half wave rectifier
ii) Full wave Rectifier

Apparatus Required

Laptop with proteus software

Theory:

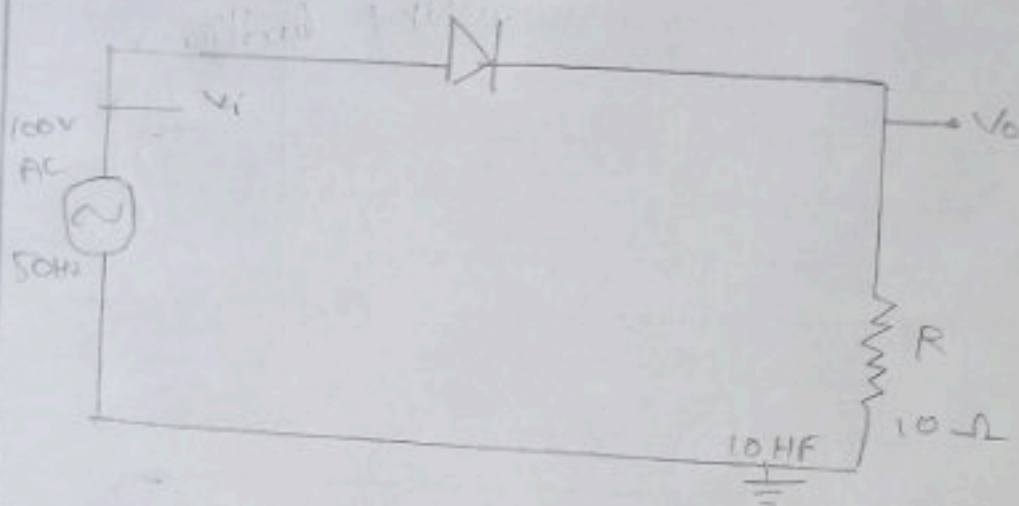
A rectifier is a circuit that converts alternating current (AC) into direct current (DC)

Half wave rectifier

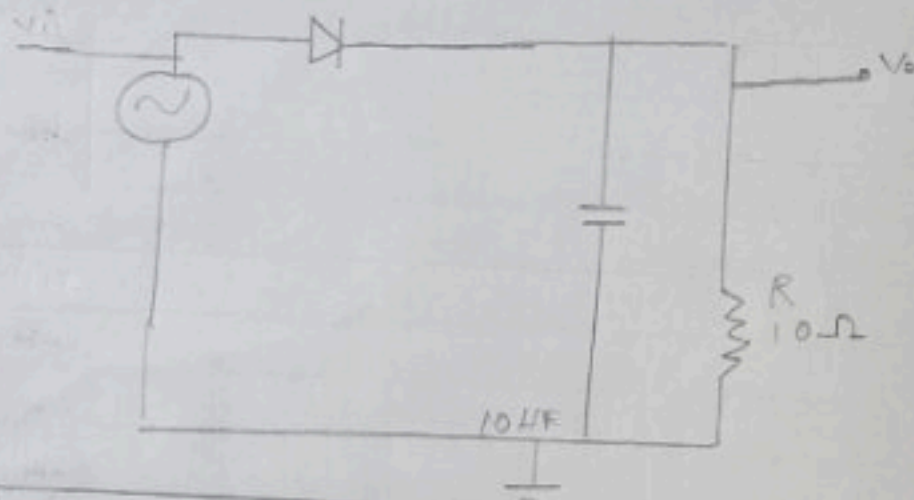
Half wave rectifier conducts only during positive half cycle. During positive half cycle the diode conducts and the output voltage is equal to the input voltage. During negative half cycle, the diode does not conduct and the output voltage is equal to zero.

Circuit Diagram

Half Rectifier
Without Filter



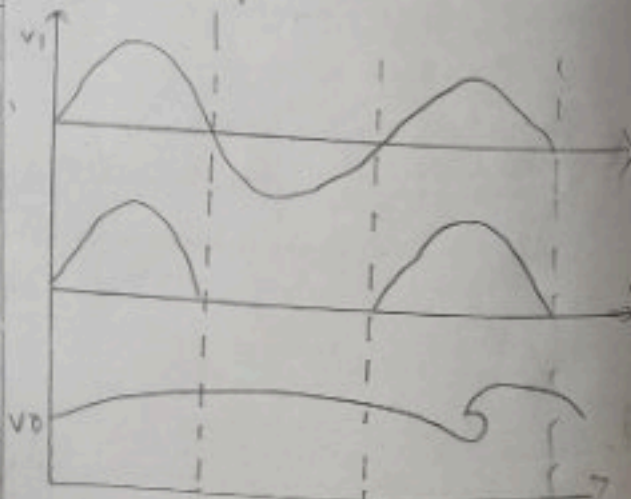
WITH FILTER



Tabulation

Input voltage		Output voltage			
		Without filter		With filter	
V_m (V)	t (ms)	V_m (V)	t (ms)	V_m (V)	t (ms)
100V	20ms	100V	100ms	100V	3ms
				80V	25ms

Model Graph



Full wave Rectifier

* Full wave rectifier conducts during both positive and negative half cycles. During positive half cycle, the output voltage is equal to input voltage during negative half cycle. The output voltage is equal to negative of the input voltage. The output of the rectifier circuit is impure DC.

In order to get pure DC the output is filtered by a capacitor filter.

Procedure

1) Drag the required components from the Proteus library

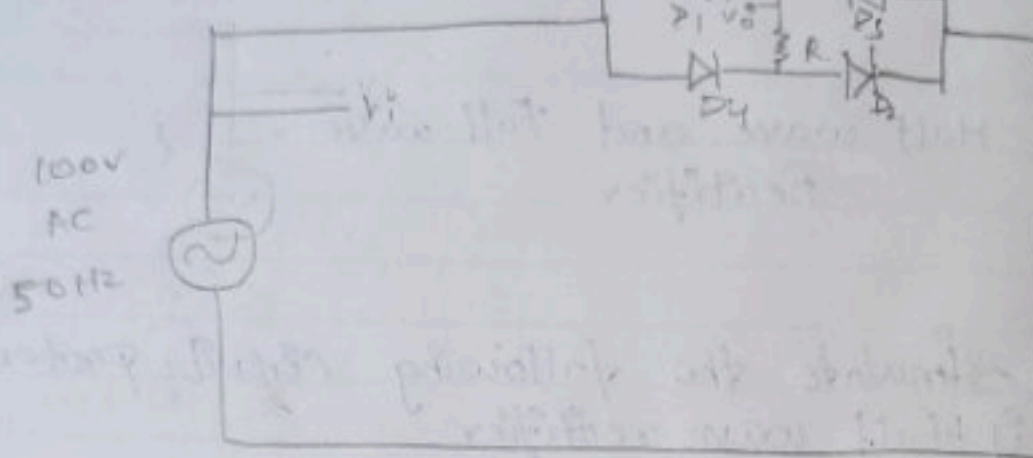
2) Connect the components as per the circuit diagram

3) Measure the input and output voltage and note in tabulations

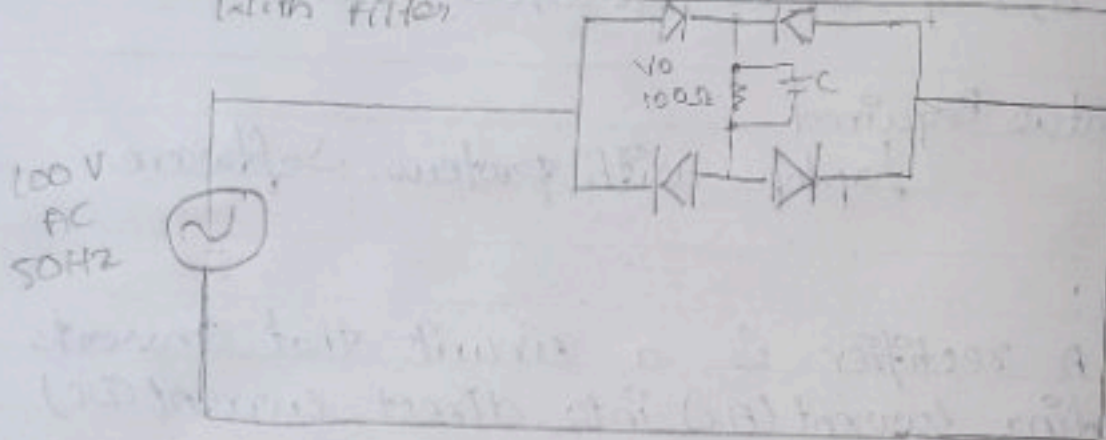
4) Draw the graph for input and output voltage.

Circuit diagram

Full wave rectifier without Filter



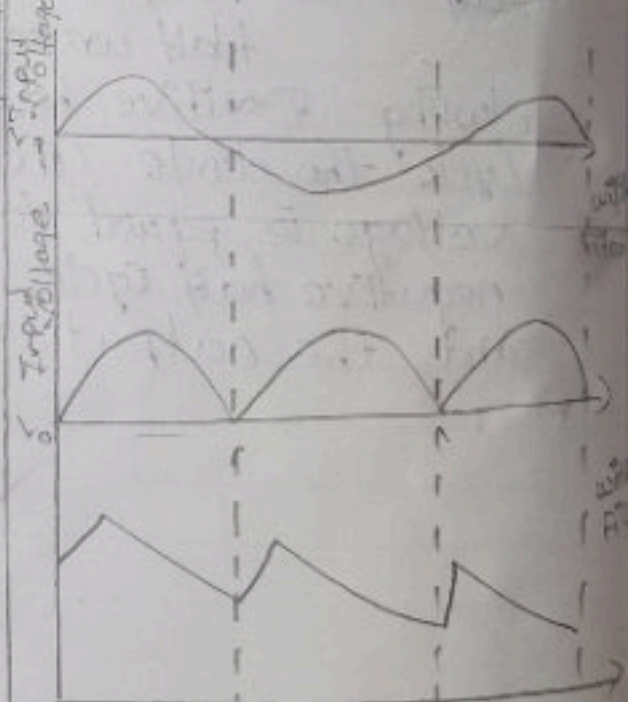
with filter



Tabulation

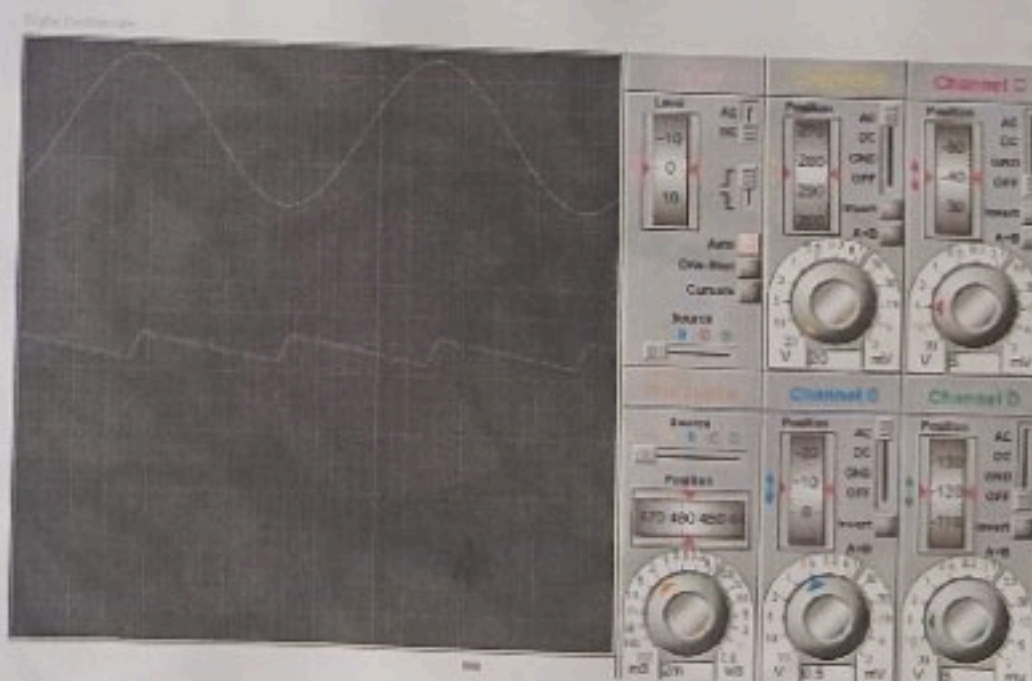
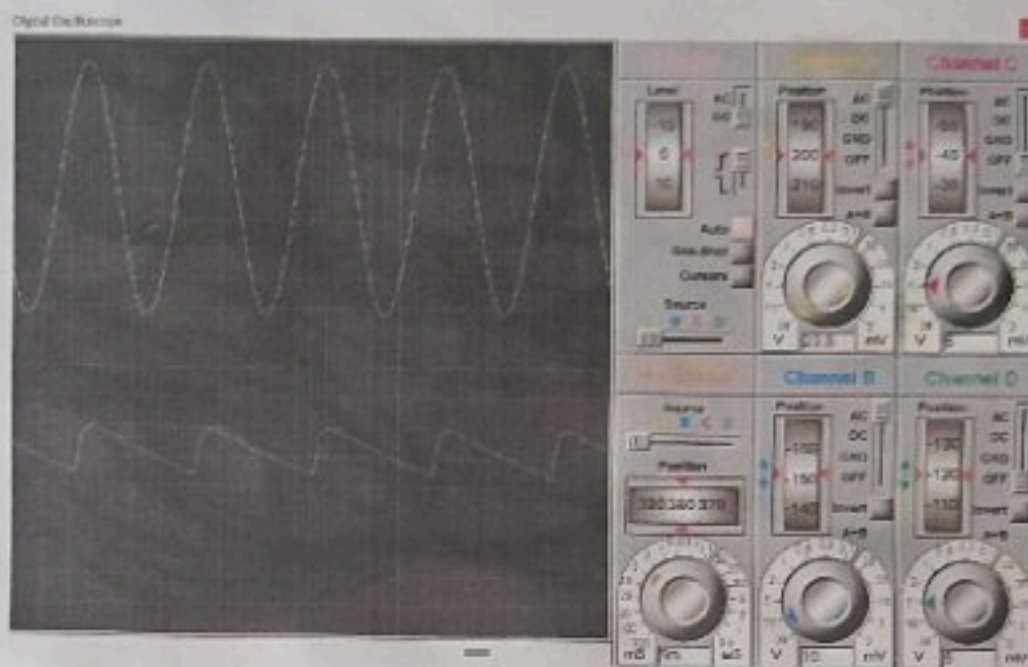
Input voltage		output voltage			
		without filter		with filter	
V_m (V)	t (ms)	V_m (V)	t_{ms}	V_{avg} (V)	t_{ms}
50V	80ms	50V	5ms	100V 5ms 40V 14ms	

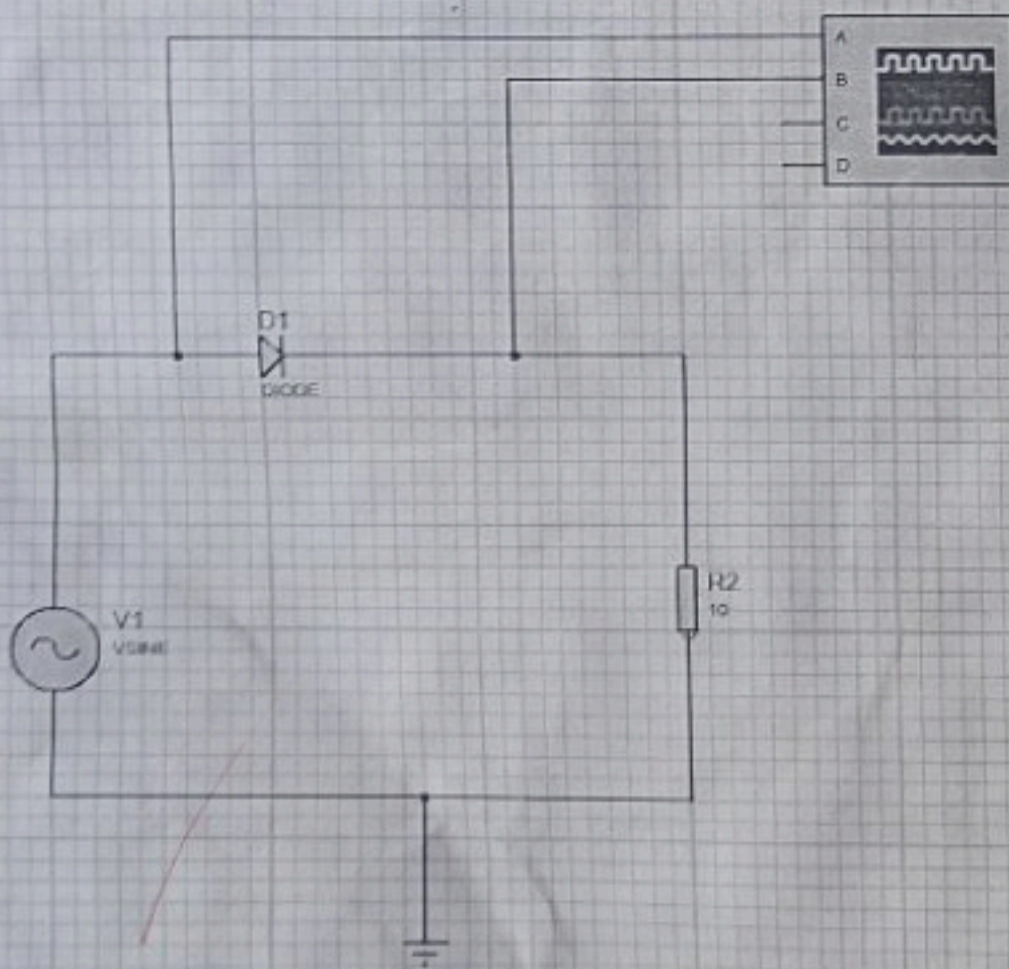
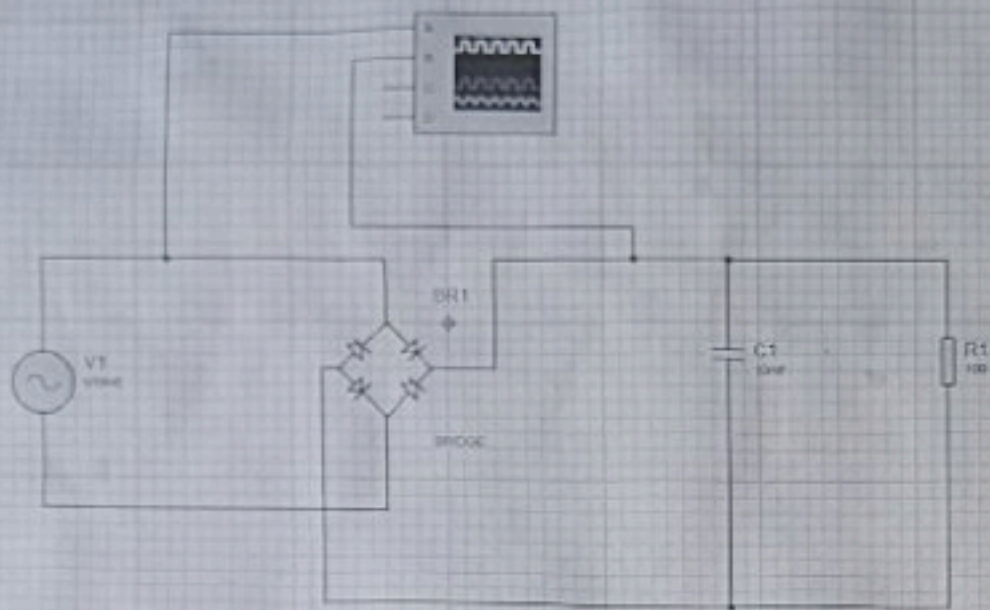
Model Graph



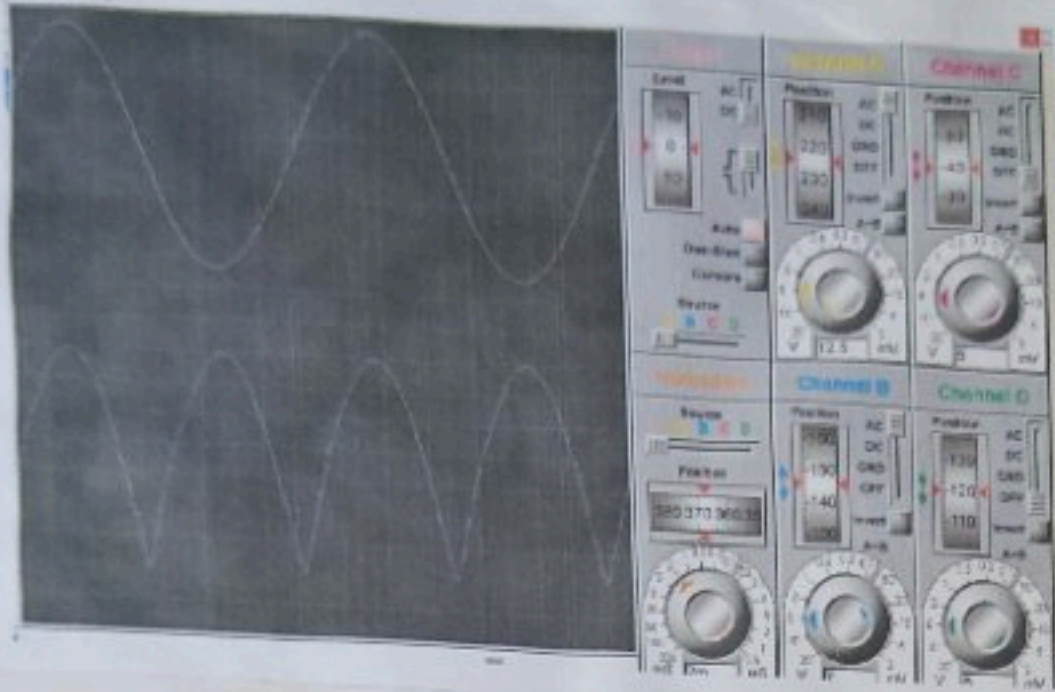
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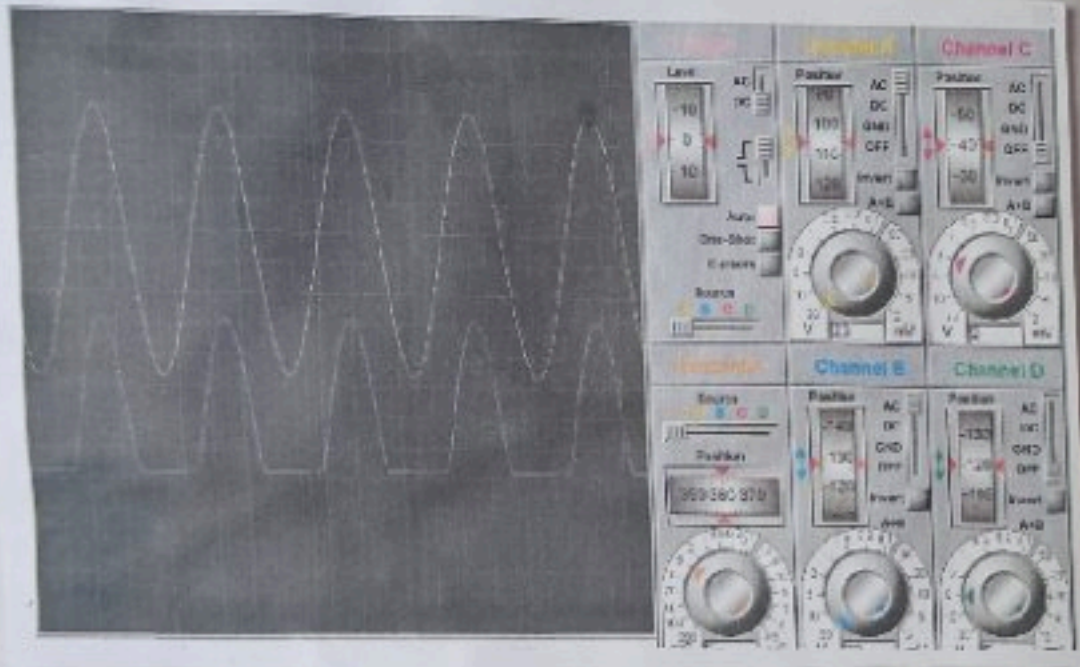


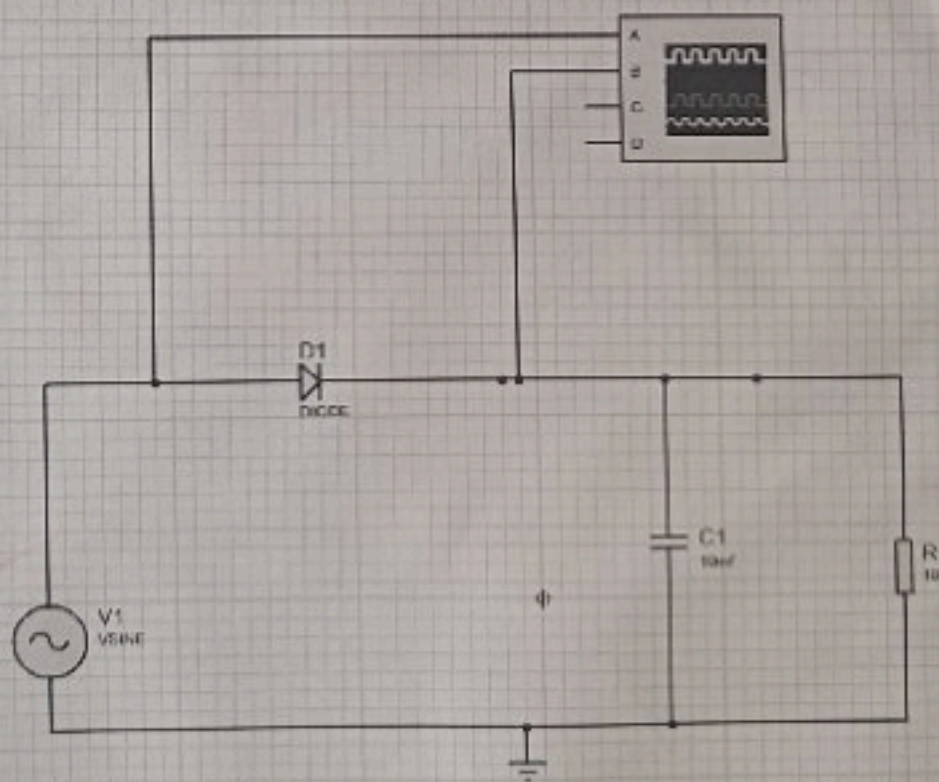
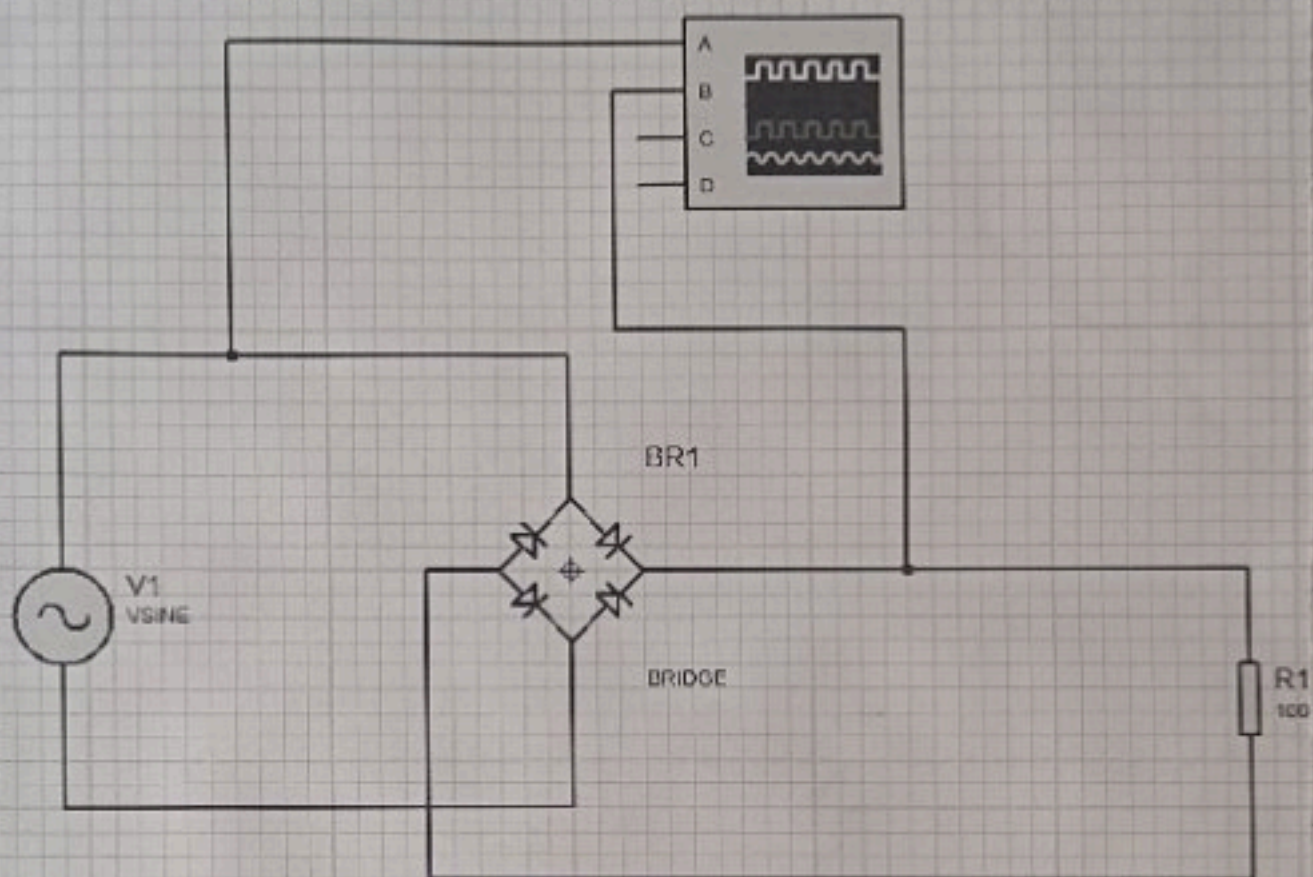


Single Channel



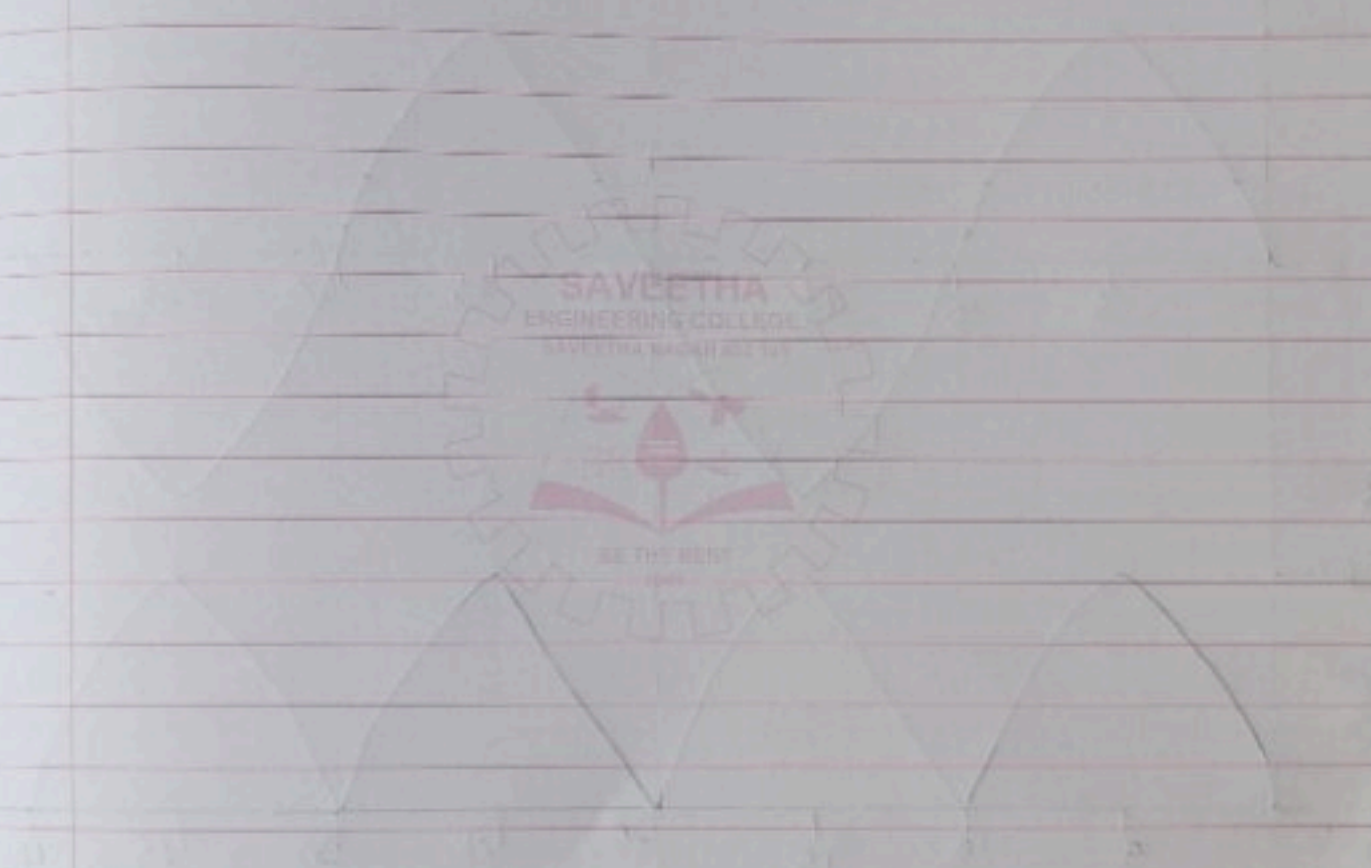
Single Channel





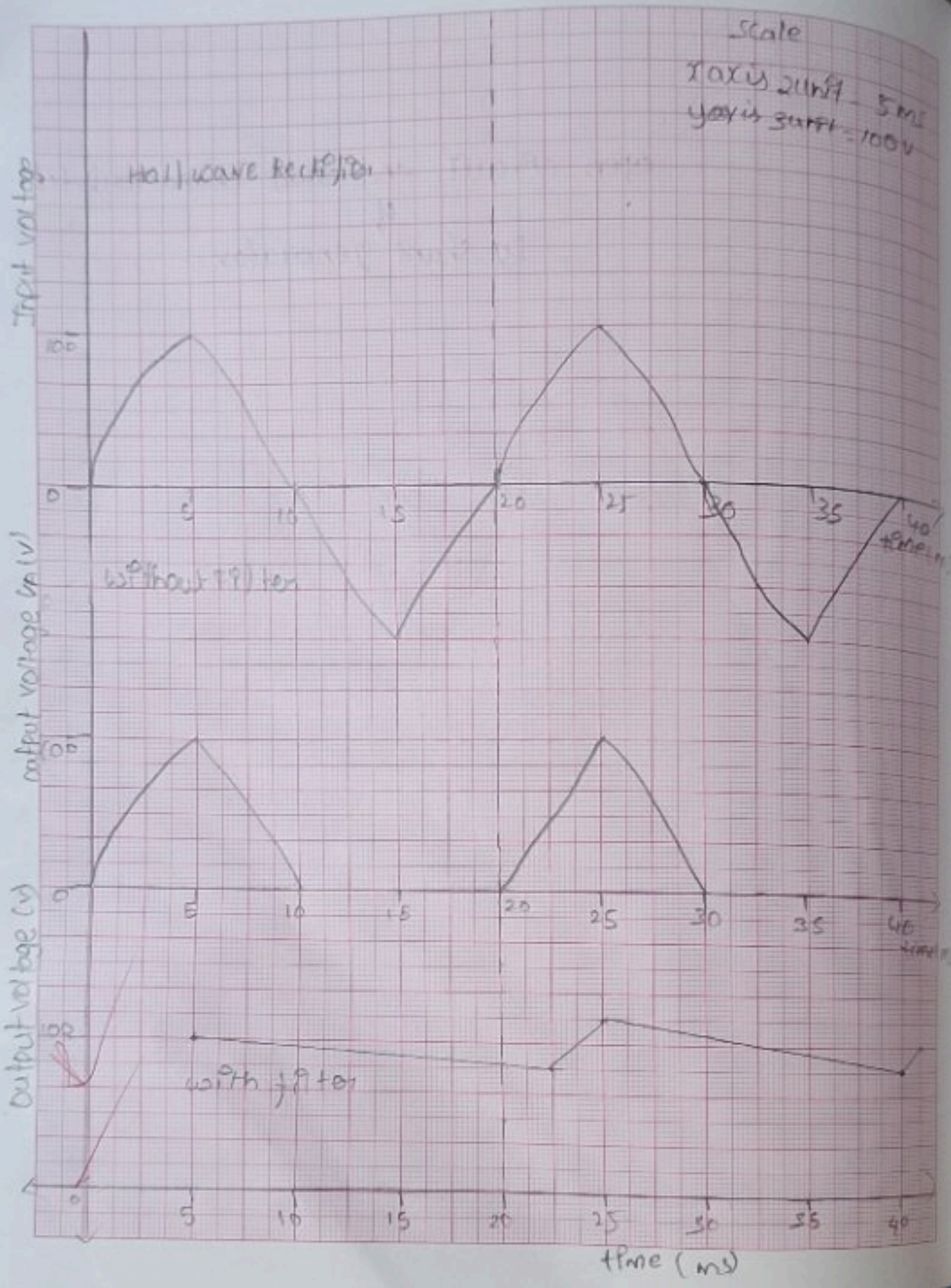
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11/6/24

Half and Full Rectifier

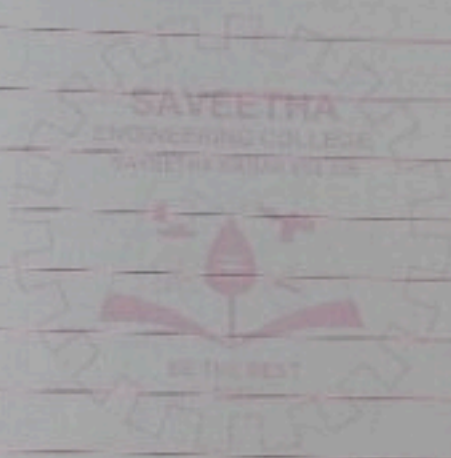


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Half and Full Rectifier



Input voltage $V_i(t)$

Full wave Rectifier

Scale
x axis unit = 5ms
y axis unit = 50V

Output voltage $V_o(t)$

